

# Credit Portfolio Risk Analysis

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**Portfolios:** Portfolio 1 and Portfolio 2

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## 1. Introduction

Credit risk modelling plays a central role in modern financial risk management, particularly in the assessment of portfolio losses and regulatory capital requirements. During periods of economic stress, defaults tend to become correlated, leading to clustered defaults and substantial portfolio losses. The Basel II Internal Ratings-Based (IRB) framework addresses this issue by incorporating asset correlation into regulatory capital calculations.

This report applies a Gaussian Threshold Model to analyse correlated default events within credit portfolios. Using Monte Carlo simulation and an analytical conditional expected loss approach, the study evaluates the impact of default correlation on portfolio loss distributions, Value-at-Risk (VaR), and Expected Shortfall (ES). The analytical conditional expected loss approach complements the default-indicator simulation by producing a smooth factor-driven approximation of portfolio losses conditional on the systematic risk factor.

## 2. Data

The analysis uses portfolio exposure data, stock returns, industry index returns, and historical default information. The datasets contain entity identifiers, Exposure at Default (EAD), Loss Given Default (LGD), booking dates, industry classifications, and probability of default indicators.

Several cleaning procedures were applied before modelling. Entity identifiers and industry labels were standardized to ensure consistent merging across datasets. Duplicate observations were removed, and all dates were converted into a unified datetime format. Only observations satisfying

$$\text{date} \geq \text{booking\_date} \tag{1}$$

were retained to avoid look-ahead bias.

Missing EAD values were replaced using average EAD values within the same size bucket. Missing LGD values were filled using industry-average or global-average LGD values. Missing probabilities of default were replaced with  $10^{-10}$  to ensure numerical stability when computing default thresholds. Missing  $\rho$  values were replaced using industry-average  $\rho$  estimates calculated from successfully estimated entities.

### 3. Methodology

#### 3.1 Literature Review

Portfolio credit risk modelling has received considerable attention in both academia and industry. Early frameworks such as CreditMetrics, CreditRisk+, and KMV Portfolio Manager were developed to model correlated defaults and portfolio loss distributions. These models emphasize the importance of portfolio-level dependence rather than analysing each obligor independently.

Among these approaches, the Gaussian latent variable framework proposed by Vasicek (2002) has become particularly influential because of its analytical tractability and its adoption within the Basel II IRB framework. Furthermore, Gordy (2003) demonstrated that the Basel II capital formula can be interpreted as a limiting form of the Vasicek single-factor model. Consequently, the Gaussian Threshold Model remains one of the most widely used approaches for portfolio credit risk measurement and regulatory capital estimation.

#### 3.2 Model Framework

This report applies a one-factor Gaussian Threshold Model to analyse correlated default risk within credit portfolios. The Gaussian Threshold framework is closely related to the Vasicek model and forms the theoretical foundation of the Basel II IRB approach to regulatory capital modelling (Vasicek, 2002; Basel Committee on Banking Supervision, 2005). The model assumes that default dependence across obligors is driven by a common systematic risk factor.

Each obligor is associated with a latent creditworthiness variable:

$$X_i = \rho_i F + \sqrt{1 - \rho_i^2} \epsilon_i \quad (2)$$

where  $F$  denotes the systematic factor,  $\epsilon_i$  represents the idiosyncratic shock, and  $\rho_i$  measures the sensitivity of obligor  $i$  to systematic risk. Both  $F$  and  $\epsilon_i$  are assumed to follow independent standard normal distributions.

Default occurs when the latent variable falls below the default threshold:

$$X_i \leq d_i, \quad d_i = \Phi^{-1}(p_i) \quad (3)$$

where  $p_i$  is the obligor's unconditional probability of default and  $\Phi^{-1}$  denotes the inverse cumulative distribution function of the standard normal distribution. This latent-variable approach is widely used in structural and portfolio credit risk modelling because it captures default clustering generated by common macroeconomic shocks (Gordy, 2003).

### 3.3 Estimation of Asset Correlation

The factor loading parameter  $\rho$  was estimated using regressions of entity stock returns on corresponding industry returns. For each obligor, the following regression was estimated:

$$R_{i,t} = \alpha_i + \beta_i R_{\text{industry},t} + \varepsilon_{i,t} \quad (4)$$

where  $R_{i,t}$  is the stock return of obligor  $i$  and  $R_{\text{industry},t}$  is the return of the corresponding industry index. The regression  $R^2$  measures the proportion of return variation explained by the common industry factor, and the asset correlation parameter was estimated as:

$$\rho_i = \sqrt{R_i^2} \quad (5)$$

This approach is economically reasonable because firms operating within the same industry are exposed to similar macroeconomic and sector-specific conditions. Therefore, the estimated  $R^2$  provides a proxy for the degree of systematic dependence within the portfolio. Similar factor-based approaches are commonly applied in portfolio credit risk modelling and Basel II correlation estimation ([Fitch Ratings, 2008](#)).

### 3.4 Monte Carlo Simulation

Monte Carlo simulation was used to generate one-year portfolio loss distributions. For each simulation, one realization of the systematic factor and one idiosyncratic shock for each obligor were drawn from standard normal distributions. Defaults were determined using the threshold condition, and portfolio losses were calculated as:

$$Loss = \sum_i EAD_i LGD_i Y_i \quad (6)$$

where  $EAD_i$  is the exposure at default,  $LGD_i$  is the loss given default, and  $Y_i$  is a binary default indicator equal to 1 if obligor  $i$  defaults and 0 otherwise. A total of 10,000 simulations were performed for each portfolio. The simulated loss distributions were then used to compute Expected Loss (EL), Value-at-Risk (VaR), and Expected Shortfall (ES).

### 3.5 Analytical Conditional Expected Loss

In addition to the default-indicator simulation, an analytical conditional expected loss approach was implemented. Instead of simulating realized defaults directly, this method computes conditional default probabilities given the systematic factor:

$$p_i(F) = \Phi \left( \frac{d_i - \rho_i F}{\sqrt{1 - \rho_i^2}} \right) \quad (7)$$

The conditional expected portfolio loss is then calculated as:

$$l(F) = \sum_i EAD_i LGD_i p_i(F) \quad (8)$$

This approach produces a smooth loss distribution driven primarily by systematic risk and allows direct comparison with the discrete default-indicator simulation. The analytical conditional expected loss framework is particularly useful for understanding the role of systematic dependence and approximating portfolio losses in highly granular portfolios (Gordy, 2003).

## 4. Results and Discussion

The comparison results show that Portfolio 2 is substantially riskier than Portfolio 1 under both simulation methods. Under the default-indicator simulation, Portfolio 1 has an Expected Loss (EL) of about 2.45 million, while Portfolio 2 has an EL of about 11.66 million. The same pattern is observed in the tail-risk measures. Portfolio 1 has a VaR at the 99% confidence level of about 26.38 million and an Expected Shortfall (ES) of about 36.30 million, whereas Portfolio 2 has a much higher VaR of about 117.29 million and an ES of about 149.39 million. This indicates that Portfolio 2 not only has higher average credit losses, but also much more severe losses under stressed scenarios.

The difference between the two portfolios is closely related to their systematic risk exposure. Portfolio 1 has an average asset correlation parameter  $\rho$  of approximately 0.446, while Portfolio 2 has a higher average  $\rho$  of approximately 0.591. According to the Basel II IRB framework, the asset correlation parameter captures the sensitivity of obligors to common systematic risk factors, where higher correlations imply stronger default clustering during economic downturns (Fitch Ratings, 2008). Therefore, the larger  $\rho$  value of Portfolio 2 helps explain why its VaR and ES values are substantially higher.

For both portfolios, the analytical conditional expected loss method produces similar EL values but lower tail-risk estimates than the default-indicator simulation. For Portfolio 1, the analytical VaR at the 99% confidence level is about 16.40 million, compared with 26.38 million under the default-indicator simulation. For Portfolio 2, the analytical VaR is about 96.75 million, compared with 117.29 million under the simulation approach. This difference arises because the analytical method smooths idiosyncratic default risk by conditioning on the systematic factor, whereas the simulation method explicitly models realized individual default events and therefore generates more discrete loss jumps.

This interpretation is consistent with Fitch Ratings' discussion of expected loss and unexpected loss. Expected loss represents the average anticipated portfolio loss, while unexpected loss reflects the variability of losses caused by systematic dependence and concentrated obligor exposures (Fitch Ratings, 2008). In concentrated portfolios, single

large defaults can significantly affect realized losses, which is more accurately reflected in default-indicator simulations.

The same pattern appears at the 99.9% confidence level. For Portfolio 1, the simulated VaR 99.9% is about 47.79 million, while the analytical VaR 99.9% is about 27.96 million. For Portfolio 2, the simulated VaR 99.9% is about 197.61 million, compared with 156.21 million under the analytical approach. This confirms that the default-indicator simulation produces more extreme tail outcomes, particularly when individual defaults create large discrete jumps in portfolio losses.

The role of the asset correlation parameter is mainly reflected in the tail of the loss distribution rather than in expected loss itself. [Fitch Ratings \(2008\)](#) similarly emphasizes that portfolios with stronger correlations exhibit larger unexpected losses because defaults tend to cluster during stress periods. Consequently, changing  $\rho$  has only a limited effect on EL, but a much stronger effect on VaR and ES.

Overall, the results suggest that Portfolio 2 has higher systematic dependence, larger expected losses, and substantially greater tail risk than Portfolio 1. The analytical conditional expected loss approach is useful for understanding the smooth effect of the systematic factor, but the default-indicator simulation better captures realized default risk in concentrated portfolios where individual defaults can strongly affect total losses. This is consistent with the argument that stress testing is essential because correlations and default clustering tend to increase during periods of financial stress ([Fitch Ratings, 2008](#)).

## 5. Conclusion

This report applied a one-factor Gaussian Threshold Model to evaluate portfolio credit risk for two credit portfolios. Using both Monte Carlo default-indicator simulation and an analytical conditional expected loss framework, the study quantified expected losses and tail risk measures under correlated default scenarios.

The results indicate that Portfolio 2 exhibits substantially higher systematic dependence and tail risk than Portfolio 1. Furthermore, the comparison between simulation-based and analytical approaches demonstrates that while the analytical method provides useful insights into systematic risk exposure, Monte Carlo simulation captures the discrete nature of defaults more accurately in concentrated portfolios.

Overall, the findings highlight the importance of asset correlation, default clustering, and portfolio concentration in credit risk management and regulatory capital assessment.

## References

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